18 August 2026

Multicomponent Molecular Editing of Polybutadiene: From Design Space to Battery Function

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Abstract

Multicomponent post-polymerization modification (PPM) reactions offer a powerful strategy for rapidly diversifying polymers, yet their potential for systematically exploring structure-property-function relationships remains largely unexplored. Herein, we report an electrophilic multicomponent molecular editing strategy that enables the construction of a polybutadiene-derived molecular design space through systematic variation of cyclic ethers and N- or O-based nucleophiles. By combining three cyclic ethers with six nucleophiles, an 18-member library was generated, allowing systematic variation of spacer architecture, polarity, and intermolecular interactions. The resulting design space was mapped across thermal and selected morphological and mobility dimensions, revealing that relatively small molecular variations induce substantial changes in macroscopic material behavior. In particular, P9 exhibited distinct nanoscale organization and proton relaxation behavior, illustrating how subtle structural variation translates into differences in morphology and molecular dynamics. Mapping these complementary landscapes guided the selection of P11, with its balanced combination of functionality and physicochemical properties, as a candidate electrolyte additive for Li-S batteries. Electrochemical evaluation revealed that P11 alters the sulfur redox response and impedance response and increases the Coulombic efficiency, resulting in a more gradual capacity decay relative to the cell containing the baseline electrolyte, while also increasing polarization and reducing the accessible capacity. Quantitative sensitivity analysis translated the multidimensional property landscapes into transferable design rules, identifying nucleophile identity as the dominant handle for functionalization degree and thermal stability, whereas the glass transition temperatures exhibited a more distributed dependence on both molecular inputs. From a broader perspective, this work establishes multicomponent molecular editing as a platform for systematically navigating multidimensional polymer design space, extracting molecular design rules, and translating molecular diversification into application-directed material selection.

Keywords

post-polymerization modification, polymer libraries, molecular design space, structure–property–function relationships, functional materials discovery, polybutadiene, Li–S battery electrolyte additives

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The bigger picture

Conventional polymer discovery often expands property space through de novo synthesis or post-polymerization modification (PPM), yet PPM strategies vary only one structural element at a time. Multicomponent molecular editing changes this paradigm by enabling multiple molecular variables to be varied systematically on a common polymer scaffold. Here, systematic variation of cyclic ether and nucleophile identity generates a chemically diverse 18-member polybutadiene library with distinct composition, thermal behavior, nanoscale organization, and molecular mobility. Rather than identifying a single structure-property relationship, the resulting landscapes reveal that different material responses are controlled by different molecular inputs: nucleophile identity dominates functionalization degree and thermal stability, whereas glass transition behavior depends more broadly on both editing components. Accordingly, this multidimensional map enabled the library-guided selection of **P11** as a promising candidate for evaluation as a **Li–S** battery electrolyte additive, demonstrating how systematic molecular diversification can be linked to application-directed material selection. Quantitative sensitivity analysis resolved the relative contributions of the molecular inputs to the different material responses, translating the experimentally mapped design space into transferable design rules. More broadly, this work positions multicomponent PPM as a platform for navigating polymer design space, resolving structure–property relationships, and guiding the selection of functional candidates for accelerated materials discovery from existing polymer scaffold.

Highlights

- Multicomponent molecular editing maps a tunable 18-member polybutadiene design space
- Nucleophile dominates functionalization and thermal stability; glass transition temperature depends on both inputs
- Subtle molecular changes reshape nanoscale organization and molecular dynamics
- Library-guided selection links polymer design space to Li–S electrolyte evaluation

Keywords

post-polymerization modification, polymer libraries, molecular design space, structure–property–function relationships, functional materials discovery, polybutadiene, Li–S battery electrolyte additives

Summary

Multicomponent post-polymerization modification (PPM) reactions offer a powerful strategy for rapidly diversifying polymers, yet their potential for systematically exploring structure-property-function relationships remains largely unexplored. Herein, we report an electrophilic multicomponent molecular editing strategy that enables the construction of a polybutadiene-derived molecular design space through systematic variation of cyclic ethers and N- or O-based nucleophiles. By combining three cyclic ethers with six nucleophiles, an 18-member library was generated, allowing systematic variation of spacer architecture, polarity, and intermolecular interactions. The resulting design space was mapped across thermal and selected morphological and mobility dimensions, revealing that relatively small molecular variations induce substantial changes in macroscopic material behavior. In particular, **P9** exhibited distinct nanoscale organization and proton relaxation behavior, illustrating how subtle structural variation translates into differences in morphology and molecular dynamics. Mapping these complementary landscapes guided the selection of **P11**, with its balanced combination of functionality and physicochemical properties, as a candidate electrolyte additive for Li-S batteries. Electrochemical evaluation revealed that **P11** alters the sulfur redox response and impedance response and increases the Coulombic efficiency, resulting in a more gradual capacity decay relative to the cell containing the baseline electrolyte, while also increasing polarization and reducing the accessible capacity. Quantitative sensitivity analysis translated the multidimensional property landscapes into transferable design rules, identifying nucleophile identity as the dominant handle for functionalization degree and thermal stability, whereas the glass transition temperatures exhibited a more distributed dependence on both molecular inputs. From a broader perspective, this work establishes multicomponent molecular editing as a platform for systematically navigating multidimensional polymer design space, extracting molecular design rules, and translating molecular diversification into application-directed material selection.

1. Introduction

Polybutadiene (**PBD**) is a widely produced synthetic polymer and constitutes a major component of commercial elastomers such as styrene-butadiene rubber (SBR), which is extensively used in tire manufacturing.[1,2] Given its large production volume and growing demand for sustainable materials management, strategies that transform polybutadiene-based materials into value-added products have become an important objective.[3,4] In this context, post-polymerization modification (PPM) offers an attractive approach, as it enables the introduction of new functionalities into existing polymer backbones without requiring the synthesis of entirely new monomers or polymers.[5] Multicomponent post-polymerization modification reactions have recently emerged as particularly powerful tools for the rapid diversification of polymer structures. In contrast to conventional sequential functionalization approaches, multicomponent reactions enable the simultaneous incorporation of several structural elements in a single transformation, thereby reducing synthetic complexity.[6] Such approaches can also potentially reduce *E*-factors, improve atom economy, and lower waste generation, particularly when reaction components such as solvents are directly incorporated into the final material.[7]

For polybutadiene and its derivatives, numerous PPM strategies have been reported to introduce polar groups,[8] reactive handles,[9] supramolecular motifs,[10] and electrochemically active functionalities.[11] While these studies have significantly expanded the accessible polybutadiene-derived materials, they have predominantly focused on individual materials. Consequently, structure-property relationships are often established on a case-by-case basis, limiting the identification of general molecular design principles. Although examples of multicomponent functionalization of polybutadiene have recently emerged,[9,12–14] the systematic exploration of multicomponent molecular design spaces and their influence on material properties remains largely unexplored.

Advances in materials discovery increasingly rely on establishing structure-property-function relationships across chemically diverse material libraries.[15,16] However, the construction of sufficiently diverse yet structurally related libraries in polymer chemistry remains synthetically demanding. We hypothesized that electrophilic multicomponent molecular editing of polybutadiene could provide a modular solution to this challenge by enabling the independent variation of two orthogonal molecular editing elements: the nucleophile and the cyclic ether. Whereas the nucleophile determines the polarity, reactivity, and intermolecular interaction profile of the introduced functionality, the cyclic ether defines the spacer architecture connecting the functionality to the polymer backbone. Together, these variables generate a chemically diverse yet structurally related molecular landscape from a common polybutadiene precursor.[17]

Herein, we report an electrophilic multicomponent molecular editing strategy that enables the rapid construction of a systematically tunable polybutadiene library. By combining six different N- and O-based nucleophiles with three cyclic ethers, an 18-member library was constructed in which both the chemical functionality and the spacer architecture were systematically varied. The resulting design space was subsequently mapped across multiple property dimensions, including thermal behavior (Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA)), nanoscale morphology (Atomic Force Microscopy (AFM), Transmission Electron Microscopy (TEM), and low-field proton magnetic resonance (^{1}H NMR) relaxometry. Comparison across these complementary property landscapes revealed that relatively subtle changes in the molecular editing components can translate into pronounced differences in higher-order material behavior. In particular, **P9** exhibited distinct nanoscale contrast and proton relaxation behavior compared with closely related library members, highlighting how molecular editing can influence both nanoscale organization and molecular dynamics. To further explore whether this library-based materials diversification could translate into application-relevant function, **P11**, was selected as a candidate electrolyte additive for Li–S batteries based on its balanced combination of chemical functionality and physicochemical properties. This selection was further motivated by the anticipated reactivity of its functional motifs: residual brominated sites could potentially undergo nucleophilic substitution with soluble polysulfides, as suggested by related organosulfur chemistry,[18] while its sulfonate ester groups provide polar sulfur- and oxygen-containing motifs with potential relevance to electrolyte/electrode interfacial processes.[19] Together, this dual-functional mechanism paves the way towards a molecular basis for investigating whether **P11** could influence polysulfide related processes and the electrochemical response of Li-S cells. Its subsequent electrochemical evaluation demonstrated how multicomponent molecular editing can connect systematic design-space exploration with the identification of functional polymer candidates. Finally, quantitative sensitivity analysis is used to extract transferable design rules from the multidimensional library, thereby distinguishing the relative influence of the cyclic ether and the nucleophile identity on functionalization and material properties.

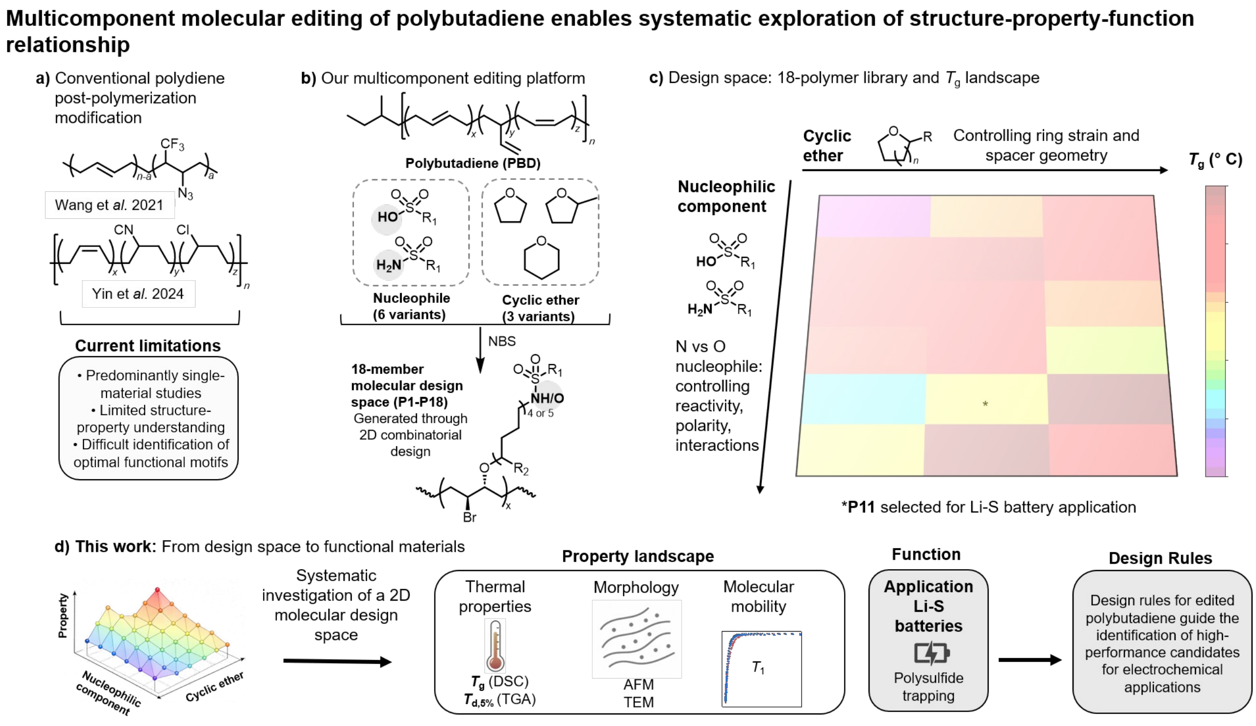


Figure 1. Construction of a multicomponent molecular editing design space for polybutadiene. (a) Current limitations of post-polymerization modification studies,[9,14] which predominantly focus on individual materials and therefore provide only limited insight into broader structure-property relationships. (b) Electrophilic multicomponent molecular editing of polybutadiene through the simultaneous incorporation of cyclic ethers and N- or O-based nucleophiles. The cyclic ether controls the spacer architecture between the polymer backbone and the introduced functionality, while the nucleophile modulates the polarity and intermolecular interaction profile of the resulting material. (c) Construction of an 18-member molecular design space by systematically varying both molecular editing components. (d) Mapping of the resulting library across multiple property dimensions, including thermal behavior (DSC, TGA), nanoscale morphology (AFM, TEM), and molecular mobility and proton relaxation behavior (low-field ^{1}H NMR relaxometry), generates structure-property landscapes that enable the identification of promising functional materials. Electrochemical evaluation of a

selected library member as an electrolyte additive for Li-S batteries provides a proof-of-concept for connecting systematic design-space exploration with functional materials discovery.

Taken together, this approach transforms a single commodity polymer into a chemically addressable materials design space, connecting molecular inputs to distinct property responses and, ultimately, electrochemical function.

2. Results and Discussion

2.1 Construction and Structural Verification of the Molecular Library

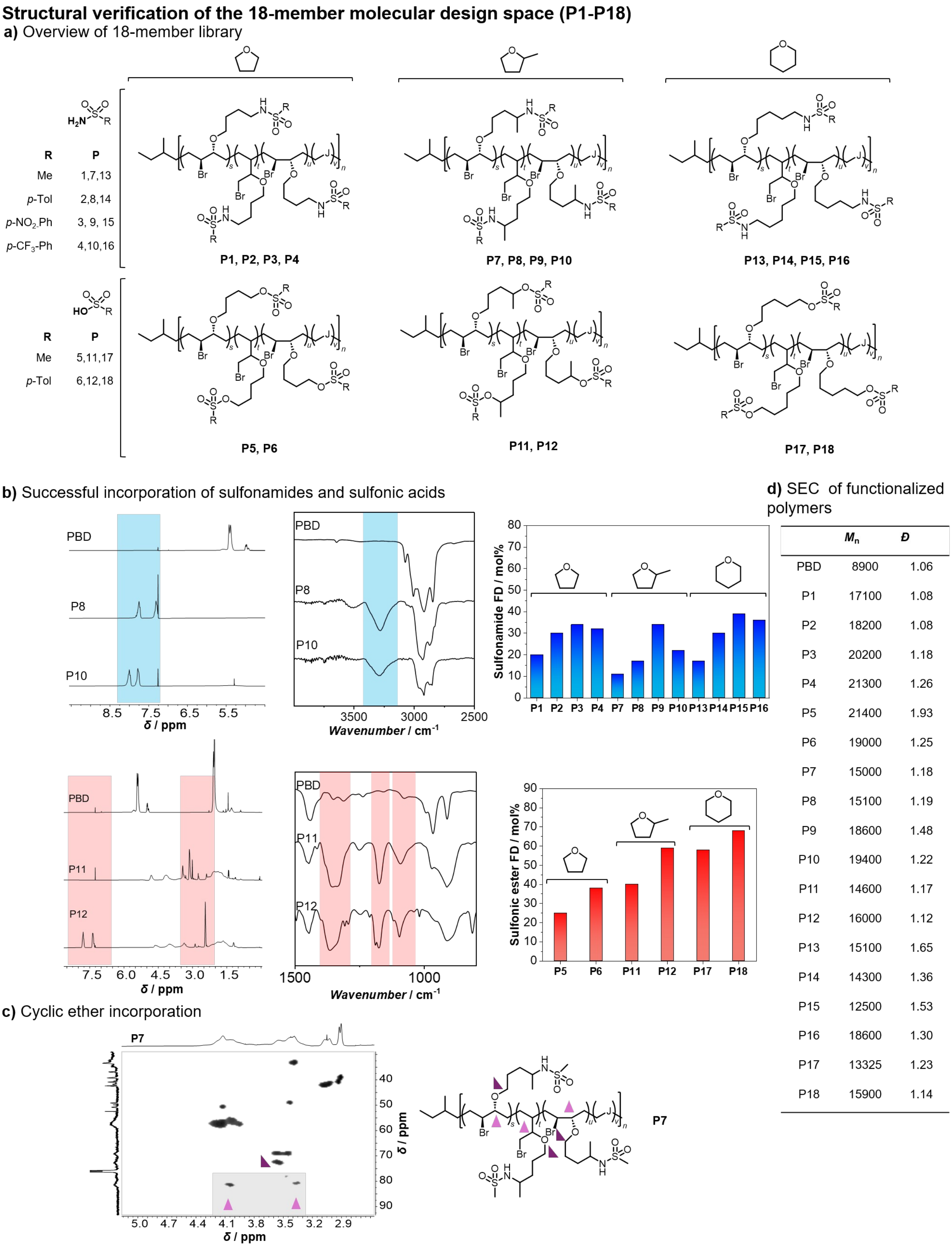
An adapted version of our previously reported electrophilic multicomponent molecular editing protocol[13] was employed to construct the molecular library (see **Supporting Information Section A.2**). Three cyclic ethers (tetrahydrofuran (THF), 2-methyltetrahydrofuran (2-MeTHF), and tetrahydropyran (THP)) were combined with six O- and N-based nucleophiles, including four sulfonamides (methanesulfonamide ($MeSO_2NH_2$), *p*-Toluenesulfonamide (*p*-$TsNH_2$), 4-Nitrobenzenesulfonamide (*p*-$NsNH_2$), and 4-(Trifluoromethyl)benzenesulfonamide (*p*-CF_3BsNH_2)) and two sulfonic acids (Methanesulfonic acid (MSA) and *p*-Toluenesulfonic acid (*p*-TsOH)), affording an 18-member polymer library (**P1**–**P18**, **Figure 2a**). This modular design enables the independent variation of two orthogonal molecular design variables: (i) the spacer architecture, defined by the cyclic ether, and (ii) the polarity and intermolecular interaction profile, governed by the nucleophile.

The successful incorporation of sulfonamide functionalities was confirmed by ^{1}H NMR and Fourier Transform Infrared Spectroscopy (FTIR). spectroscopy. Representative ^{1}H NMR spectra of **P8** and **P10** (**Figure 2b)** have shown the expected aromatic resonances arising from the introduced sulfonamide substituents; FTIR spectra have displayed the characteristic N–H stretching vibration at approximately at 3300 cm^{-1} together with the expected S=O and C–O stretching vibrations (**Figure 2b** and **Supporting Information Figures S55-S72**). Structural assignments for all library members were further supported by ^{13}C NMR spectroscopy (cf. **Supporting Information Figures S2-S54** for all NMR spectra). Quantitative ^{1}H NMR (qNMR) spectroscopy was used to estimate the functionalization degree (FD) (cf. **Supporting Information Section A.4** for calculation details). The FD (mol%) depended on both the cyclic ether and the nucleophile, highlighting that the two molecular editing components jointly influence the final polymer composition. Across all three cyclic ether-derived subsets, sulfonamide FDs ranged from 20–34 mol% for the THF series (**P1**–**P4**), 11–34 mol% for the 2-MeTHF series (**P7**–**P10**), and 17–39 mol% for the THP series (**P13**–**P16**) **(Table S1**). Within each cyclic ether-derived subset, the degree of incorporation varied non-monotonically with sulfonamide identity, highlighting the pronounced influence of the nucleophile on functionalization efficiency. The observed differences in the FD can be rationalized by the electrophilic multicomponent editing mechanism, which proceeds through initial bromonium ion formation, followed by ring opening by the cyclic ether and subsequent nucleophilic trapping of the resulting oxonium intermediate.[17] Because both molecular editing components participate directly in this reaction sequence, variation of either component can influence the overall functionalization outcome. The cyclic ether may influence ring-opening energetics and intermediate stability, whereas the nucleophile governs trapping of the oxonium intermediate through differences in its steric and electronic properties. The non-monotonic incorporation trends observed for the sulfonamide series indicate that functionalization is controlled by the interplay of nucleophile reactivity, intermediate stability, steric accessibility, and polymer solubility rather than by a simple electronic effect.

The formation of the sulfonate ester derivatives was similarly confirmed by NMR and FTIR spectroscopy (with representative spectra of **P11** and **P12** shown in **Figure 2b)**. As expected, the sulfonate ester derivatives lacked N-H stretching vibrations characteristic of the sulfonamide derivatives, while retaining the characteristic S=O absorption bands. Quantitative qNMR/internal-standard analysis, performed in an analogous manner to the sulfonamide-decorated polymers, has revealed consistently higher FDs for the sulfonate esters compared to the corresponding sulfonamides. For all three cyclic ethers, the *p*-toluenesulfonic acid-derived polymer exhibited higher incorporation than its methanesulfonic acid-derived counterpart, yielding 25 and 38 mol% for **P5** and **P6**, 40 and 59 mol% for **P11** and **P12**, and 58 and 68 mol% for **P17** and **P18**, respectively (**Table S1**). In addition, the sulfonate ester FD increased systematically across the cyclic ether series, following the trend THF < 2-MeTHF < THP for both sulfonic acid nucleophiles. Within the proposed reaction pathway, this trend may reflect differences in oxonium-ion stability and nucleophile accessibility, together with polymer-reaction-medium interactions associated with the respective cyclic ethers. Notably, the THP-derived sulfonate ester polymers **P17** and **P18** exhibited the highest FD values within the entire library, with **P18** reaching 68 mol%. Consequently, these results show that the multicomponent editing strategy provides not only structural diversity but also a tunable incorporation landscape governed by both nucleophile class and cyclic ether identity.

Successful incorporation of the cyclic ether component was further verified by two-dimensional (2D) NMR spectroscopy. The Heteronuclear Single Quantum Coherence (HSQC) spectrum of **P7** (**Figure 2c**) displayed the expected correlation between the characteristic carbon resonance at ~80 ppm and adjacent proton resonances of the incorporated ether unit at 4.1 and 3.3 ppm. Furthermore, ^{13}C NMR confirmed the incorporation of **THF**, **2-MeTHF**, and **THP-**derived spacer units throughout the library (**SI section C.1**).

Finally, Size-Exclusion Chromatography (SEC) analysis in THF provided complementary evidence for changes in the hydrodynamic behavior following molecular editing relative to the polybutadiene precursor. Compared to pristine **PBD** (M_n = 8900 g mol^{-1}, $Đ$ = 1.06), all edited polymers exhibited increased apparent molar masses (**Figure 2d** and **Supporting Information Table S1**). Some polymers exhibited broader apparent molecular weight distributions in THF; however, this behavior is primarily attributed to changes in hydrodynamic volume and polymer-solvent interactions arising from the introduced polar functionalities. For instance, **P3** exhibited an apparent dispersity value ($Đ$) of 1.53 in THF, whereas analysis in *N*,*N*-Dimethylacetamide (DMAc) yielded a significantly lower value of $Đ$ = 1.14 (**Supporting Information Figure S89**), highlighting the pronounced influence of the eluent and polymer-solvent interactions on the SEC-derived values. Hence, the combined NMR, FTIR, HSQC, and SEC analyses confirm the successful construction of the 18-member polymer library. The molecular editing strategy thus provides access to a chemically diverse yet structurally related polymer library for systematically investigating structure-property-function relationships.



	M_n	$Đ$
PBD	8900	1.06
P1	17100	1.08
P2	18200	1.08
P3	20200	1.18
P4	21300	1.26
P5	21400	1.93
P6	19000	1.25
P7	15000	1.18
P8	15100	1.19
P9	18600	1.48
P10	19400	1.22
P11	14600	1.17
P12	16000	1.12
P13	15100	1.65
P14	14300	1.36
P15	12500	1.53
P16	18600	1.30
P17	13325	1.23
P18	15900	1.14

Figure 2. Construction and structural verification of the 18-member molecular design space (P1-P18). (a) Overview of the library generated by combining three cyclic ethers (**THF**, **2-MeTHF**, and **THP**) with four sulfonamides (**$MeSO_2NH_2$**, ***p*TsNH₂**, ***p*NsNH₂**, and ***p*CF_3BsNH_2**) and two sulfonic acids (**MSA** and ***p*TsOH**). (b) Successful incorporation of sulfonamides and sulfonate esters is confirmed by representative ^{1}H NMR and FTIR spectra together with the functionalization degree (FD), determined by quantitative ^{1}H NMR spectroscopy (the calculation is detailed in **the SI Section A.4**). (c) Verification of cyclic ether incorporation by HSQC NMR spectroscopy (exemplified for **P7**). (d) SEC analysis demonstrating increased apparent molar masses relative to pristine polybutadiene.

2.2 Thermal Property Landscape of the Polymer Library

The influence of molecular editing on the thermal behavior of **PBD** was evaluated by differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA). The glass transition temperatures (T_gs) and thermal degradation temperatures at 10% mass loss ($T_{d,10\%}$) were determined for all 18 library members (cf. **Supporting Information Figures S101-S103** and **S105-S119**, respectively). The resulting thermal property landscapes are summarized in **Figure 3**.

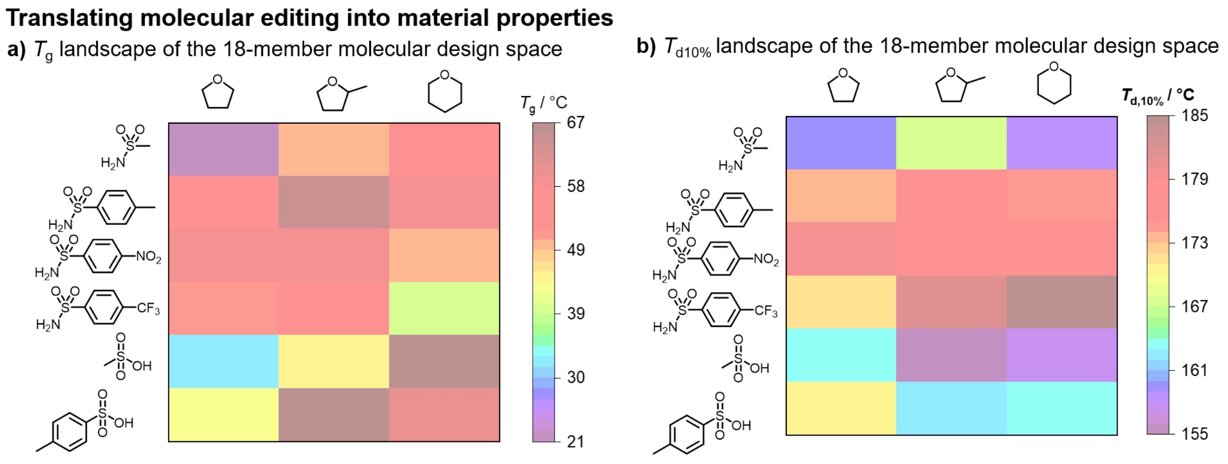


Figure 3. Thermal property landscapes of the 18-member polymer library (a) Glass transition temperatures (T_gs) and (b) degradation temperature at 10% mass loss ($T_{d,10\%}$). Rows correspond to the incorporated nucleophiles (**$MeSO_2NH_2$**, **$pTsNH_2$**, **$pNsNH_2$**, **pCF_3BsNH_2**, **MSA**, and **pTsOH** from top to bottom), while columns represent cyclic ether used during molecular editing. The left, middle, and right columns correspond to the **THF**-derived polymers (**P1**-**P6**), **2-MeTHF**-derived polymers (**P7**-**P12**), and **THP**-derived polymers (**P13**-**P18**), respectively. T_g values range from 21 to 67 °C and $T_{d,10\%}$ values range from 155 to 185 °C, respectively.

Molecular editing produced a broad thermal-property landscape across the polymer library, with T_g values ranged from 21 °C (**P1**) to 67 °C (**P12** and **P17**), while $T_{d,10\%}$ values varied between 155 °C (**P11**) and 185 °C (**P16**). These results demonstrate that relatively small variations in the molecular editing components (i.e., the cyclic ether and nucleophile) can substantially influence the thermal behavior of the edited polymers. The T_g landscape revealed a clear dependence on the cyclic ether-derived spacer, although the magnitude and direction of this effect depended on the nucleophile identity. For several nucleophile families, the transition from THF- to 2-MeTHF- and THP-derived structures resulted in higher glass transition temperatures. In the methanesulfonamide series, T_g increased from 21 °C for **P1** to 50 °C for **P7** and 53 °C for **P13**, respectively. Similarly, the MSA-derived sulfonate ester series showed T_g values of 32, 45, and 67 °C for **P5**, **P11**, and **P17**, respectively. The *p*-toluenesulfonate series exhibited comparable behavior, with T_g values of 43, 67, and 60 °C for **P6**, **P12**, and **P18**, respectively. Although the increase was not strictly monotonic in the latter series, both the 2-MeTHF- and THP-derived polymers exhibited higher T_g values than the corresponding THF-derived polymer. These variations likely reflect differences in side-chain architecture, steric environment, segmental mobility, and intermolecular interactions introduced by the cyclic ether-derived spacers. However, the influence of spacer structure was not universal across all nucleophile families. For example, in the p-nitrobenzenesulfonamide derivatives **P3**, **P9**, and **P15** displayed T_g values of 58, 58, and 50 °C, respectively, while the *p*-CF_3-substituted sulfonamide derivatives **P4**, **P10**, and **P16** exhibited values of 51, 53, and 40 °C, respectively. The absence of a universal monotonic relationship between spacer structure and T_g therefore indicates that the thermal response cannot be attributed to spacer identity alone. Rather, T_g appears to reflect the combined effects of spacer architecture, nucleophile-derived functionality, degree of incorporation, steric environment, and intermolecular interactions. Thus, these results demonstrate that variation of both molecular editing components provides access to a broad range of thermal properties from the same parent polymer scaffold.

The thermal stability landscape showed a different dependence on molecular structure. All library members displayed degradation temperatures above 155 °C. The highest thermal stabilities were observed for several aromatic sulfonamide-containing polymers, including **P3**, **P9**, **P10**, and **P16**, with $T_{d,10\%}$ values of 179, 178, 181, and 185 °C, respectively. The enhanced thermal stability of these materials may be associated with their aromatic substituents and differences in intermolecular interactions. In contrast, the sulfonate ester-containing polymers generally displayed lower thermal stabilities. The MSA-derived sulfonate ester polymers **P5**, **P11**, and **P17** exhibited $T_{d,10\%}$ values of 163, 155, and 157 °C, respectively, while **P6**, **P12**, and **P18** showed values of 170, 162, and 163 °C, respectively.

Notably, the trends observed for T_g and $T_{d,10\%}$ were not directly correlated. For example, **P12** and **P17**, both with T_g values of 67 °C, exhibited only moderate thermal stability, with $T_{d,10\%}$ values of 162 and 157 °C, respectively. Conversely, **P16** displayed the highest thermal stability of the library ($T_{d,10\%}$ = 185 °C) despite a comparatively low T_g of 40 °C. These observations indicate that segmental mobility and thermal degradation are governed by distinct, although potentially overlapping, structural factors and further illustrate the multidimensional nature of the molecular design space.

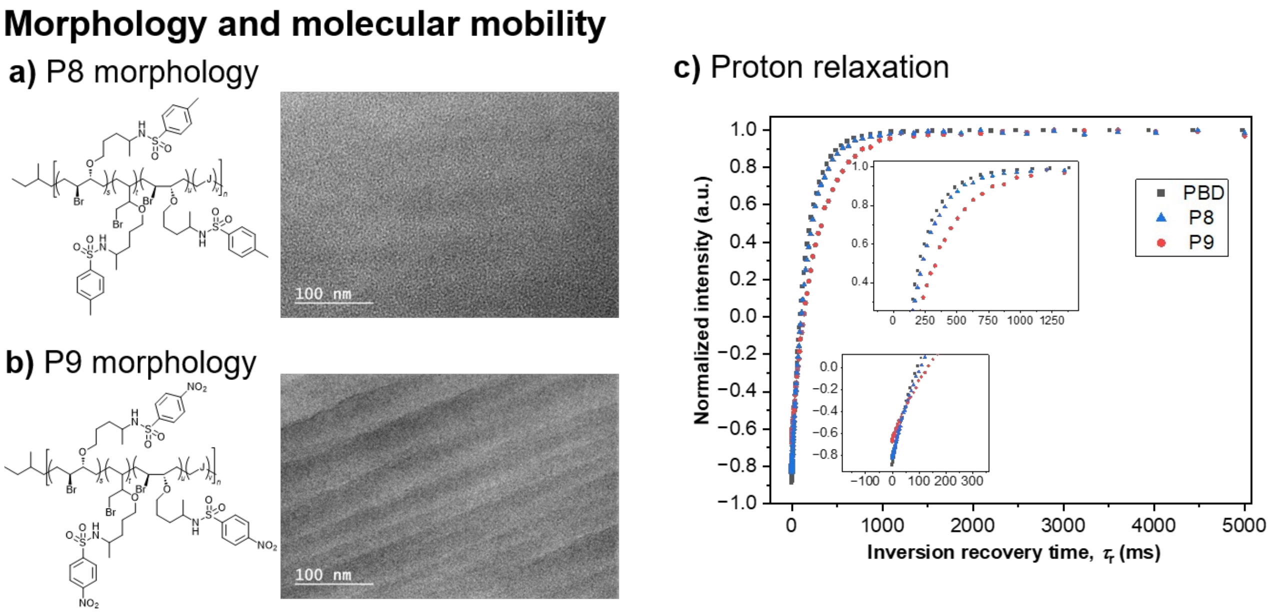
To evaluate whether the observed thermal behavior was primarily governed by the extent of functionalization, the T_g was plotted against the FD determined by qNMR/internal-standard analysis and mass-balance calculations (**Supporting Information Figure S104**). Only a weak global correlation was observed (R^2=0.17). For example, polymers with similar functionalization levels (such as **P1** and **P10**) displayed substantially different T_g values, whereas **P18,** which exhibited the highest estimated incorporation, did not show the highest T_g.

The thermal property landscape revealed that systematic variation of both cyclic ether and nucleophile identity enables substantial tuning of the thermal properties of the molecularly edited polymers. More importantly, the absence of simple one- parameter correlation emphasizes that these properties emerge from the combined influence multiple molecular design variables, including spacer architecture, nucleophile identity, functionalization level, and intermolecular interactions. The respective multidimensional behavior supports a library-based approach for identifying structure–property relationships in molecularly edited polybutadienes.

2.3 The Influence of Molecular Editing on Nanoscale Organization

To investigate whether molecular editing influences nanoscale organization and molecular mobility, selected library members were investigated by transmission electron microscopy (TEM), atomic force microscopy (AFM), and low-field ^{1}H NMR relaxometry. **P9** was initially selected because of the high polarity of the *p*-

nitrobenzenesulfonamide functionality, the expected favorable electron contrast in TEM, and its relatively high FD of 34 mol% (**Table S1**, entry 9). **P8** and **P9** were then selected for direct comparison because both contain the same **2-MeTHF**-derived spacer but differ only in the sulfonamide substituent. Notably, TEM analysis of bulk samples prepared by microtomy revealed markedly different nanoscale contrasts for the two polymers (**Figure 4a,b**).



d) Mobility fingerprint

Polymer	$T_{1,\mathrm{eff}}$ (ms)	A_{SE}/A_{FID}	Apparent FID signal loss (%) relative to SE	A_{MSE}/A_{FID}	Apparent FID signal loss (%) relative to MSE
PBD	149 ± 1	0.98	n.d.	0.96	n.d.
P8	172 ± 1	1.12	11	1.12	11
P9	262 ± 2	1.22	18	1.30	23

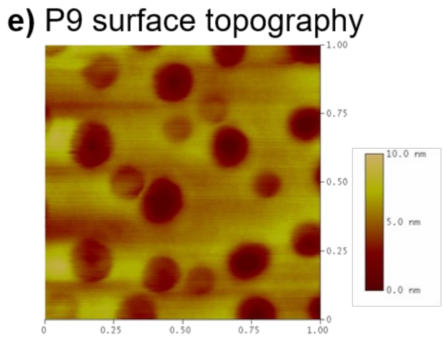


Figure 4. Morphology and molecular mobility of selected molecularly edited PBD derivatives. (a,b) Chemical structures and representative TEM images of **P8** and **P9**, respectively. (c) Normalized ^{1}H inversion–recovery profiles of pristine **PBD**, **P8**, and **P9** measured by low-field NMR relaxometry. Insets highlight the early- and intermediate-time regions of the recovery curves. (d) Comparative mobility fingerprint summarizing $T_{1,\mathrm{eff}}$, solid-echo (SE)-to-FID and magic-sandwich echo (MSE)-to-FID amplitude ratios, and the corresponding apparent FID signal losses. Apparent signal-loss values are used as comparative pulse-sequence-dependent descriptors (**Supporting Info Section C.8** for details regarding the methodology). (e) Representative AFM height image of **P9**, revealing heterogeneous surface topography and nanoscale domains; Feret-diameter analysis gave an average apparent domain diameter of 207 nm.

Whereas **P8** displayed comparatively weak contrast, **P9** exhibited pronounced, regularly spaced contrast features with a characteristic separation of approximately 30 nm (**Figure S120**). Although the precise origin of this contrast cannot be assigned unambiguously from TEM alone, the comparison indicates that variation of the incorporated sulfonamide functionality may substantially influence nanoscale organization. Importantly, the same qualitative trend was observed for the **THF**-derived analogues **P2** and **P3** (**Figures S121–S122**). As for **P8** and **P9**, the *p*-nitrobenzenesulfonamide-containing polymer **P3** exhibited more pronounced nanoscale contrast than its *p*-toluenesulfonamide counterpart **P2**, although the contrast in **P3** was somewhat weaker than that observed for the corresponding **2-MeTHF**-derived polymer **P9**. The recurrence of this qualitative trend across both cyclic-ether-derived spacer architectures suggests that the identity of the sulfonamide substituent contributes to the observed nanoscale organization, while the spacer structure may modulate its extent. The stronger contrast associated with the nitro-substituted polymers may partly reflect their higher heteroatom content and associated electron contrast, but differences in intermolecular interactions and organization may also contribute; these effects cannot be distinguished from the present TEM data alone. Motivated by the particularly distinct TEM morphology of **P9**, its surface topography was further examined by AFM (**Figure 4e**). The height image revealed a heterogeneous thin-film surface containing discrete nanoscale domains, with Feret-diameter analysis giving an average apparent domain diameter of approximately 207 nm. This length scale is substantially larger than the approximately 30 nm periodic contrast observed by TEM and is therefore not interpreted as representing the same structural feature. Rather, the two techniques provide complementary views of differently prepared material states: TEM was performed on sections representative of the bulk polymer, whereas AFM probes the surface morphology of a deposited thin film. The difference between the bulk and thin-film morphologies may arise from confinement and processing effects during film formation. Accordingly, the characteristic length scales obtained by TEM and AFM are not directly correlated. Together, however, the measurements demonstrate that **P9** exhibits structural heterogeneity across different nanoscale length scales and material geometries.

To assess whether the observed morphological differences are accompanied by changes in proton dynamics, **P8** and **P9** were further compared with pristine **PBD** by low-field ^{1}H NMR relaxometry (**Figures 4c,d and S124**). Monoexponential analysis of the inversion–recovery profiles yielded effective longitudinal relaxation times, $T_{1,\mathrm{eff}}$ of 149 ± 1, 172 ± 1, and 262 ± 2 ms for pristine **PBD**, **P8**, and **P9**, respectively (**the fitting method and all calculation details are detailed in Supporting Info Section C.8**). The substantially longer $T_{1,\mathrm{eff}}$ observed for **P9** demonstrates that incorporation of the *p*-nitrobenzenesulfonamide functionality produces a distinctly different proton relaxation environment compared with both pristine **PBD** and **P8**. Because T_1 depends on molecular motions occurring on the relevant relaxation timescale, these values are treated as comparative effective relaxation parameters rather than as direct measures of polymer rigidity. Moreover, comparison of direct free-induction decay (FID), solid-echo (SE), and magic-sandwich echo (MSE) signals provided a complementary probe of rapidly dephasing proton contributions. Relative to the direct FID, **P8** exhibited an apparent signal loss of approximately 11% when referenced to either SE or MSE, whereas the corresponding losses for **P9** increased to 18% and 23%, respectively (**Figure 4d** and **Supporting Info Section C.8** for calculation details). No positive apparent signal loss was observed for pristine **PBD**. Because these values depend on the pulse sequence and experimental time window, they are not interpreted as absolute rigid- or mobile-proton fractions. Instead, the combined $T_{1,\mathrm{eff}}$ and FID/SE/MSE measurements provide a comparative mobility fingerprint, with **P9** displaying both the most distinct longitudinal relaxation behavior and the largest contribution from rapidly dephasing proton environments. Hence, the TEM, AFM, and low-field ^{1}H NMR results demonstrate that molecular editing extends beyond chemical diversification to influence both nanoscale organization and proton dynamics. The recurrence of related TEM trends across the **THF-** and **2-MeTHF-**derived polymer pairs further indicates that these effects are not restricted to a single spacer architecture, revealing a multidimensional molecular design space in which the identities of both the sulfonamide substituent and cyclic-ether-derived spacer can influence higher-order material properties. These structure-dependent differences further motivate investigating a selected library member as a functional material for a technologically relevant application, linking molecular editing to performance in a real-world materials context.

2.4 Application of **P11** as a functional Electrolyte Additive for Li–S Batteries

P11 was selected as a representative candidate based on a combination of chemical functionality and material properties. In particular, **P11** contains aliphatic sulfonate ester functionalities, providing sulfur- and oxygen-rich polar motifs with potential relevance to electrolyte/electrode interfaces. Sulfur-containing functionalities are well-established electrolyte-active motifs in lithium batteries, where they can participate in interphase formation, and related sulfonate-containing additives have also been shown to influence polysulfide speciation and interfacial processes in Li–S systems.[20] This literature precedent motivated our evaluation of the sulfonate-ester-functionalized polymer as an electrolyte additive, without presupposing an analogous mechanism for the polymer-bound functionality. At the same time, **P11** occupies an intermediate position within the molecular library: its FD of 40 mol% lies between, for example, the other aliphatic sulfonic ester functionalized derivatives **P5** (25 mol%) and **P17** (58 mol%), and its T_g of 45 °C similarly falls within the intermediate range of the investigated materials. **P11** additionally exhibits a relatively narrow molecular-weight distribution ($Đ$=1.17) and a $T_{d,10\%}$ of 155 °C, well above the typical operating temperature range of Li–S cells. These characteristics, therefore, made **P11** a suitable candidate for assessing whether the molecularly edited PBD platform could translate its tunable structural and physical properties into function in a technologically relevant Li–S battery environment.

The need to investigate alternative electrolyte additives arises from the limitations of lithium nitrate ($LiNO_3$), a widely used additive in Li–S batteries. $LiNO_3$ contributes to the formation of a protective interphase on the lithium-metal anode, thereby mitigating parasitic reactions between soluble polysulfides and lithium and improving Coulombic efficiency. However, this passivation strategy addresses the consequences of polysulfide migration at the anode rather than the underlying dissolution and diffusion of polysulfides themselves. Also, $LiNO_3$ is progressively consumed during cycling, and its decomposition may contribute to gas-generating side reactions.[21] Therefore, there is interest in exploring alternative electrolyte additives that may mitigate polysulfide-related parasitic reactions without relying on $LiNO_3$. In this study, **P11** was evaluated as a potential additive in a $LiNO_3$-free ether-based electrolyte. As mentioned above, **P11** contains sulfonate ester and brominated functional groups (**Figure 5a**), which could potentially interact with electron-rich polysulfide species. Such interactions could affect polysulfide solubility, transport, or conversion in Li-S cells. However, the nature and strength of these interactions were not directly investigated in the present study. Accordingly, **P11** was evaluated experimentally as a potential electrolyte additive.

As described in the **Supporting Information** section **B.9**, **P11** completely dissolves in the baseline electrolyte (1M LiTFSI in DME/DOL, 1:1 v/v) and exhibits pronounced concentration-dependent behavior (**Figure 5b**). At a concentration of 1.5 wt%, homogeneous solutions were obtained, whereas increasing the concentration to 3.5 wt% resulted in gelation after approximately two days. At 6 wt%, gelation occurred within minutes after preparation. The spontaneous, concentration-dependent gelation observed at 3.5wt% and 6.0wt% **P11** in the absence of any applied electrochemical potential suggests that **P11** actively alters the physical state of the electrolyte at elevated concentrations. The molecular origin of this gelation, however, cannot be established from this observation alone. One plausible explanation is **P11** induced cationic ring-opening polymerization of DOL, considering the precedent of Lewis acid-induced polymerization reported for $Al(OTf)_3$, LiDFOB, and BF_3 in ether-based electrolytes.[22, 18] Alternatively, gelation could involve direct interactions or reactions involving the functionalities of **P11,** or a combination of processes. Further studies, including solvent-specific controls (DME-only vs. DOL-only and FTIR analysis, are required to distinguish between **P11**-induced DOL polymerization, direct nucleophilic substitution involving the brominated groups, or the coexistence of both mechanisms.

To explore the electrochemical response of the **P11**-containing electrolyte at the sulfur-electrode environment while excluding contributions from lithium metal, cyclic voltammetry was performed using a symmetric S||S cell (**Figure 5ci**). A relatively low current response was observed between approximately -0.7 V to +0.7 V, followed by a pronounced increase in current outside this voltage range. Because the measurement was conducted in a two-electrode configuration without a reference electrode, the contributions of the individual electrodes and the origin of the observed currents cannot be unambiguously resolved. Accordingly, this experiment provides an exploratory assessment of the overall electrochemical response of the P11-containing electrolyte in the sulfur electrode environment, rather than a mechanistic evaluation of sulfur conversion or **P11** stability. To investigate the influence of **P11** on the lithium-metal/electrolyte interface, EIS measurements were performed using Li||Li symmetric cells containing the baseline electrolyte and 1.5wt%_**P11** (**Figure 5cii**). Before lithium plating/stripping, the cells exhibited different impedance responses, with apparent arc diameters of approximately 130 Ω for the baseline electrolyte and 180 Ω for the **P11**-containing electrolyte, as estimated from the Nyquist plots. After plating/stripping, both spectra showed larger and similar apparent arc diameters of approximately 215 Ω. Thus, under the investigated conditions, **P11** initially altered the impedance response of the Li||Li cell, whereas no clear additional impedance contribution associated with **P11** was observed after plating/stripping. However, EIS alone cannot determine whether **P11** decomposes or participates in the formation of the solid electrolyte interphase (SEI) at the lithium electrode.

To evaluate whether **P11** fulfills its proposed role as an electrolyte additive, its impact on Li-S cell behavior was investigated using complementary electrochemical techniques. Due to the functional groups present in **P11**, the additive was hypothesized to interact with soluble polysulfide species and thereby affect sulfur conversion and polysulfide migration. However, the present electrochemical measurements do not directly demonstrate these molecular interactions. Cyclic voltammetry of the full Li-S cells shown in **Figure 6ai** reveal that the addition of 1.5wt%_**P11** substantially alters the sulfur redox response compared with the baseline cell. Specifically, the redox peaks become broader and the separation between anodic and cathodic peaks increases, indicating a more polarized electrochemical response in presence of P11. The Nyquist plots shown in Figure 6**aii** additionally reveal pronounced differences in the impedance response. Both cells exhibit a comparable high-frequency intercept of approximately 6 Ω, indicating comparable overall ohmic resistance under the investigated conditions. In contrast, the apparent arc diameter, estimated visually from the Nyquist plots, increases from approximately 130 Ω for the baseline cell to approximately 315 Ω for the **P11**-containing cell prior to cycling, indicating that **P11** modifies the overall impedance response. Nonetheless, the molecular origin of this increased impedance cannot be resolved by EIS alone. After six CV cycles, the apparent arc diameter for the **P11**-containing cell decreases to approximately 140 Ω, indicating substantial change of the impedance response during the initial cycling. This change could reflect improved electrode wetting, activation of the sulfur cathode, and/or restructuring of an additive-derived surface layer, though EIS alone cannot distinguish between these possibilities

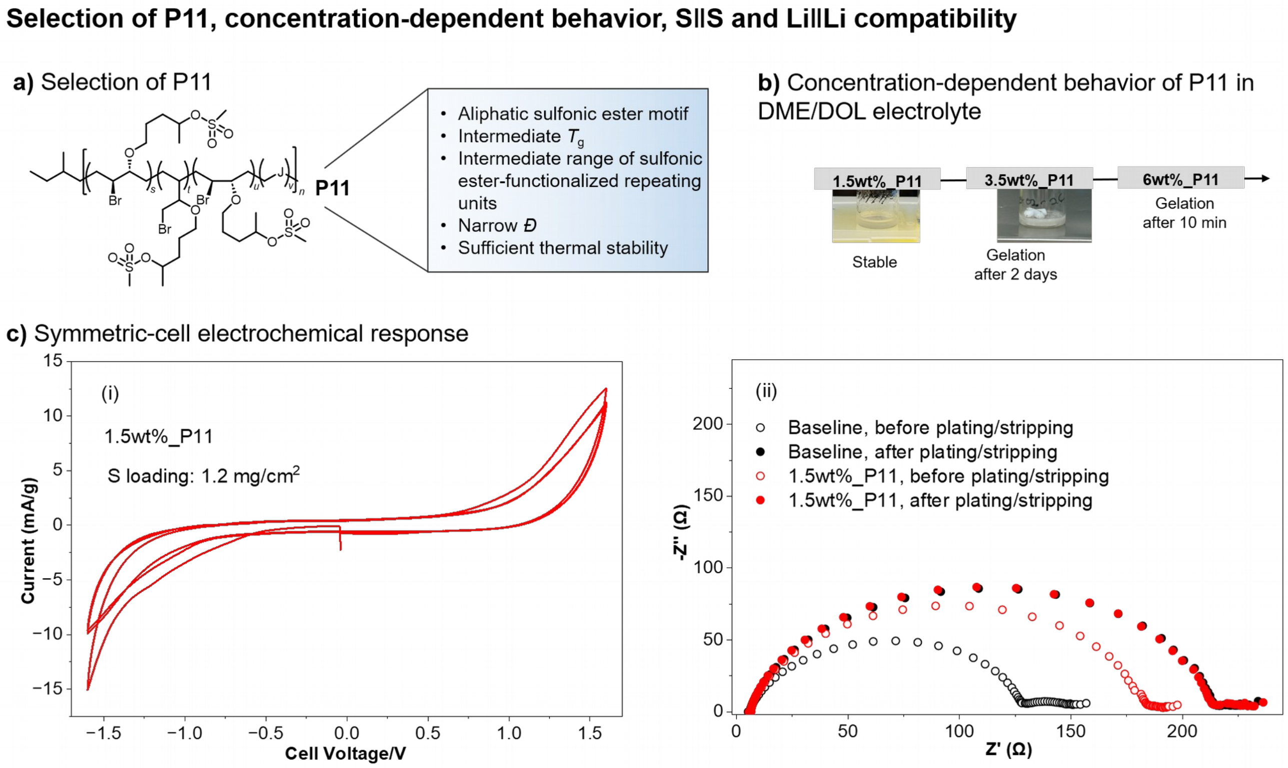


Figure 5. Selection of **P11**, concentration-dependent electrolyte behavior, electrochemical response in S‖S and Li‖Li symmetric cells. (a) Selection of **P11** based on its balanced combination of sulfonate ester functionality, narrow molar mass distribution, intermediate T_g, and sufficient thermal stability. (b) Concentration-dependent behavior of **P11** in 1 M LiTFSI DME/DOL electrolyte, showing a homogeneous electrolyte at 1.5 wt% and gel formation at higher concentrations. (c) Electrochemical response in symmetric cells: (i) cyclic voltammetric response of **P11**-containing electrolyte in a symmetric S‖S cell and (ii) impedance spectra of Li||Li cells before and after three full cycles of plating/stripping.

The influence of the altered polarization and impedance response is also reflected in the cycling performance shown in **Figure 6aiii**. The **P11**-containing cell exhibits a lower initial discharge capacity of approximately 275 mAh/g_s than the baseline cell. However, long-term electrochemical behavior reveals a different trend. The cell containing the baseline electrolyte undergoes continuous capacity decay accompanied by a relatively low CE of approximately 78%, a pattern typically associated with persistent polysulfide shuttling and irreversible loss of active material in Li–S cells. In contrast, the P11-containing cell exhibits a considerably slower capacity-fading rate together with a higher CE of ~93%, although a gradual decline is observed during extended cycling. The improved CE and enhanced capacity retention are consistent with a reduction in parasitic processes associated with soluble polysulfide species over extended cycling, although the present data do not directly establish suppression of polysulfide migration or stabilization of soluble sulfur species as the underlying cause. Altogether, these indicate an electrochemical trade-off: while the additive initially increases polarization and the apparent impedance response, it is associated with higher Coulombic efficiency and more gradual capacity decay over extended cycling

The electrochemical response of 1.5wt%_**P11** containing Li–S cells also depends on electrolyte volume, as shown by comparing cells with electrolyte volumes of 100 µL [electrolyte-to-sulfur (E/S) ≈ 49 µL/mg] and 70 µL (E/S ≈ 36 µL/mg). The cyclic voltammogram (**Figure 6bi**) shows broad, low-magnitude Faradaic features for the cell containing 100 µL, whereas the cell containing 70 µL exhibits sharper, higher-current peaks at ~2.3 V and ~2.0 V vs. Li^+/Li. Because the GFA separator and electrode architecture were identical between conditions, and both volumes are expected to provide adequate cell wetting, differences in electrode/electrolyte contact are unlikely to account for the sharper response at the lower electrolyte volume. Instead, the different responses may reflect changes in the concentration and transport of soluble polysulfide species within the available electrolyte volume: at 100 µL, the larger electrolyte reservoir could favor greater spatial distribution and dilution of soluble intermediates, whereas at 70 µL, these species may remain more concentrated within the smaller volume. Such differences in polysulfide distribution and transport could contribute to the broader CV response at 100 µL and the sharper features at 70 µL. On this basis, the CV differences are consistent with an electrolyte-volume-dependent concentration and transport-distribution effect and should not, by themselves, be interpreted as evidence for faster intrinsic sulfur redox kinetics at the lower electrolyte volume.

This interpretation is further supported by the Nyquist plot data **(Figure 6bii**). For cells containing 100 µL and 70 µL of electrolyte, corresponding to E/S ratios of approximately 49 and 36 µL/mg, respectively, the high-frequency intercept is similar (~6 Ω), indicating comparable ohmic resistance. However, the overall apparent impedance diameter is smaller for the cell containing 70 µL (~50 Ω at OCV, ~25 Ω after CV) than for the cell containing100 µL (~125 Ω at OCV, ~80 Ω after CV). These values were read directly from the Nyquist plots and represent the total impedance response of the cell, comprising the high-frequency intercept together with the overlapping features at lower frequencies; no equivalent-circuit fitting was performed, and the values are therefore not assigned to any individual process such as charge transfer or interfacial film formation. Consistent with this lower apparent ~~interfacial~~ impedance, the 70 µL cell also exhibits a higher rate capability, delivering ~375 mAh/g_s at 2 C (**Figure 6biii**). Importantly, this improved rate response does not by itself establish faster intrinsic sulfur redox kinetics and may instead reflect electrolyte-volume-dependent differences in interfacial transport and soluble-species distribution. However, the advantage observed at lower electrolyte volume is not sustained during prolonged cycling. Long-term cycling (**Figure 6biv**) shows that the cell containing 70 µL, despite its favorable early-stage rate performance, undergoes progressive CE decline (toward ~78%), whereas the 100 µL cell sustains a higher Coulombic efficiency (~83%) beyond 345 cycles. A single abrupt drop in capacity is observed near cycle 300 for the 70 µL cell; as no corresponding change is evident in the surrounding cycles or in the voltage response, this isolated point is attributed to a measurement artefact rather than to a genuine loss of cell function, and cycling of this cell was not continued beyond this point. This decline in CE at the lower E/S ratio is consistent with increased susceptibility to parasitic reactions during extended cycling at reduced electrolyte volume. Thus, these results suggest an electrolyte-volume-dependent trade-off: reducing the electrolyte volume is associated with a lower initial impedance response and higher rate capability but with earlier capacity loss during prolonged cycling under the investigated conditions.

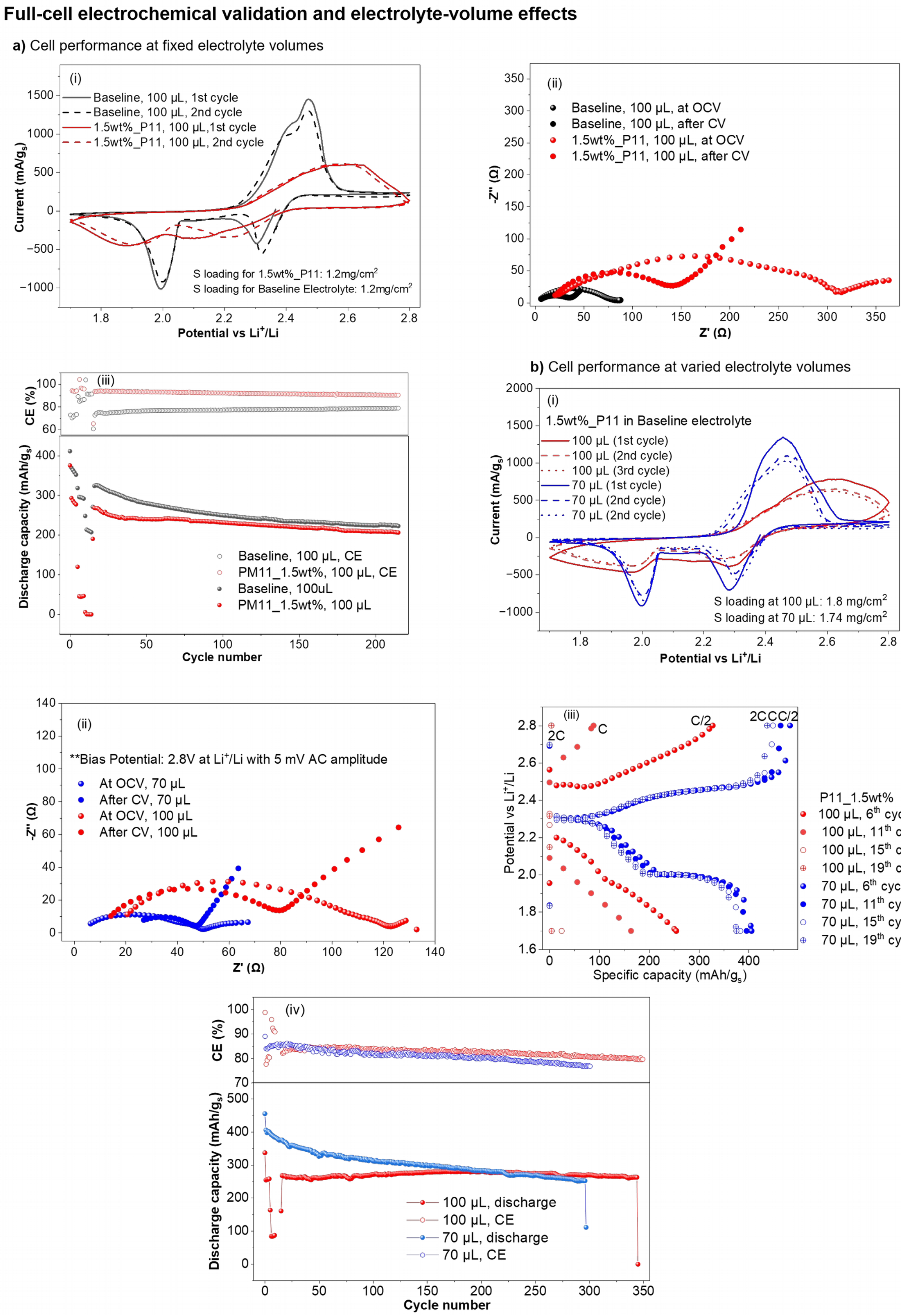


Figure 6. Full-cell electrochemical performance of **P11**-containing Li-S cells and electrolyte-volume effects. (a) Comparison of the baseline electrolyte and the electrolyte containing 1.5 wt%_**P11**: (i) cyclic voltammetric response, (ii) Nyquist plots before and after CV cycling and (iii) long-term cycling performance. (b) Effect of electrolyte volume on cells containing 1.5 wt% **P11**: (i) cyclic voltammetric response, (ii) Nyquist plots before and after CV cycling, (iii) rate performance and corresponding voltage profiles, and (iv) long-term cycling performance.

Finally, to extract general design rules from the multidimensional library, the relationships between molecular inputs, resulting material states, and material responses were integrated into a unified framework (**Figure 7**). Variation of cyclic ether and nucleophile identity defines the molecular input space (**Figure 7, panel 1**), generating distinct material states characterized by functionalization degree (FD) and, for selected representatives, molecular mobility and nanoscale morphology (**Figure 7, panel 2**). These material states translate into distinct thermal-property landscapes, while the evaluation of **P11** as an electrolyte additive provides a selected proof of application-relevant function (**Figure 7, panel 3**). Importantly, the FD, T_g, and $T_{d,10\%}$ landscapes are non-congruent (**Figure 7, panels 2A and 3**), demonstrating that different material characteristics respond differently to the two molecular editing variables. FD varies predominantly with nucleophile identity, T_g exhibits a more distributed dependence on both molecular inputs, and $T_{d,10\%}$ is strongly nucleophile-dependent but essentially insensitive to cyclic ether identity. Quantitative sensitivity analysis substantiated these trends (**Figure 7, panel 4; Supporting Information, Section C.9**): nucleophile identity accounted for 67.6% and 82.6% of the observed variation in FD and $T_{d,10\%}$, respectively, whereas cyclic ether identity contributed only 12.5% and 0.5% (**Table S2**). In contrast, T_g showed more comparable contributions from cyclic ether (23.9%) and nucleophile identity (29.1%), together with substantial remaining or non-additive variation (**Table S2**). The selected mobility and morphology measurements further illustrate how this multidimensional response extends beyond the library-wide thermal properties. Under constant 2-MeTHF conditions, changing the nucleophile from *p*-$TsNH_2$ in **P8** to *p*-$NsNH_2$ in **P9** increased $T_{1,eff}$ from 172 to 262 ms and increased the apparent loss of rapidly relaxing proton signal. Similarly, the selected TEM comparisons revealed nucleophile-dependent differences in nanoscale organization that recurred across both the THF- and 2-MeTHF-derived polymer pairs, while the available data indicated a comparatively weaker effect of cyclic ether identity (**Figure 7, panels 2B, 2C, and 4; Supporting Information, Section C.9**). Accordingly, these non-equivalent sensitivity profiles demonstrate that polymer composition, thermal behavior, molecular dynamics, and nanoscale organization cannot be predicted from a single molecular descriptor. Rather, multicomponent molecular editing provides a systematic means to identify which molecular inputs govern specific material properties within a common polymer scaffold. The resulting design rules therefore transform the library from a collection of individual polymers into a multidimensional structure–property–function map that can guide the selection of materials with targeted property combinations.

3. Conclusions

In summary, we have demonstrated that electrophilic multicomponent molecular editing provides a modular strategy for constructing chemically diverse polymer libraries from a common polybutadiene precursor. By systematically varying both cyclic ether and nucleophile identity, an 18-member library, with distinct compositional, thermal, morphological, and molecular-mobility characteristics was generated. Mapping the resulting response landscapes revealed that relatively small molecular variations can induce substantial changes in material behavior and, importantly, that the different responses are not governed uniformly by the molecular input variables. Quantitative sensitivity analysis translated these landscapes into molecular design rules, revealing that nucleophile identity constitutes the dominant molecular handle for FD and $T_{d,10\%}$, whereas T_g displayed a more distributed and non-additive dependence on both cyclic ether and nucleophile identity.

Selected mobility and morphology measurements further indicated substantial nucleophile-dependent changes in molecular dynamics and nanoscale organization. Consequently, these non-equivalent sensitivity profiles demonstrate that no single molecular descriptor, including FD, is sufficient to predict the full material-property landscape. This library-based analysis further guided the selection of **P11** as a candidate electrolyte additive with a balanced combination of chemical functionality and physicochemical properties. Subsequent preliminary evaluation in Li-S cells showed that **P11** alters the sulfur redox behavior and overall impedance response and increases the Coulombic efficiency, resulting in more gradual capacity decay relative to the baseline cell. These effects are accompanied by increased polarization and reduced accessibility, revealing an electrochemical trade-off under the investigated conditions. Rather than establishing an optimized electrolyte additive, these results provide a proof of concept that systematic exploration of a molecularly edited polymer library can be connected to the identification and evaluation of application-relevant functional candidates

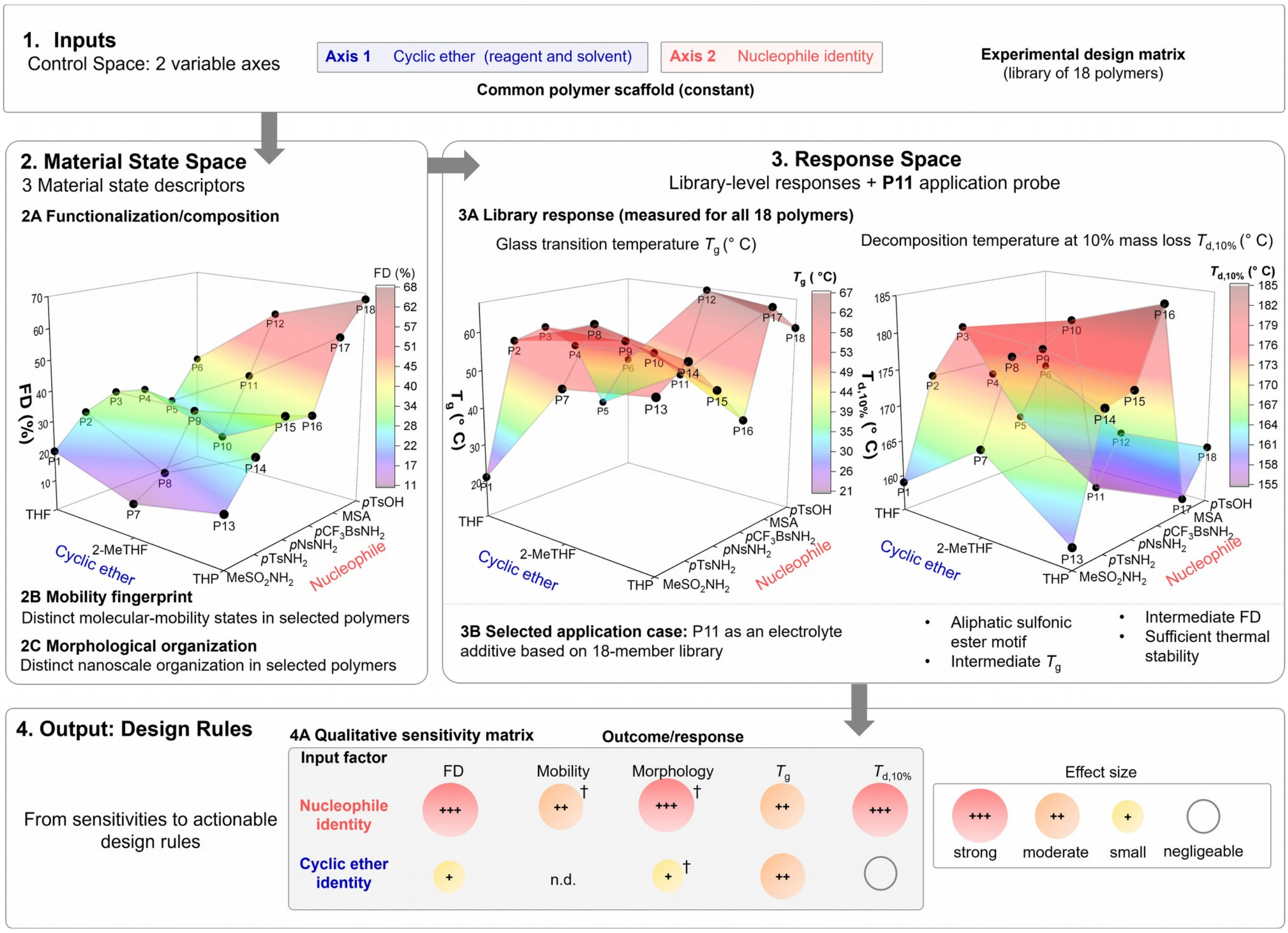


Figure 7. Molecular design space, material-state descriptors, response landscapes, and resulting design rules across the 18-member polymer library. (**1**) Inputs: the control space is defined by two molecular editing variables, cyclic ether and nucleophile identity while maintaining a common polymer scaffold. (**2**) Material State Space: molecular editing generates distinct material states characterized by functionalization degree (FD), molecular mobility, and nanoscale morphology; mobility and morphology were evaluated for selected representative polymers. (**3**) Response Space: library-wide T_g and $T_{d,10\%}$ surfaces reveal distinct thermal-property landscapes, while **P11** serves as a selected application probe for evaluating the library as a source of functional electrolyte additives. (**4**) Output-Design Rules: the relationships between molecular inputs and material characteristics are summarized in a sensitivity matrix, revealing distinct sensitivities to nucleophile and cyclic ether identity and thereby providing molecular design rules for tuning different material characteristics. **O**, negligible; +, weak; ++, moderate; +++, strong sensitivity; **n.d.**, not determined; **†** qualitative assessment based on selected polymers. Calculation and classification of the sensitivity matrix are detailed in **Section C.9** of the **Supporting Information**.

Thus, at a broader level, this work establishes multicomponent molecular editing not only as a versatile synthetic methodology but also as a strategy for navigating multidimensional polymer design spaces and extracting molecular design rules from structure-property-function relationships. By systematically varying multiple molecular parameters within a common scaffold, this approach transforms polymer diversification from the generation of individual derivatives into the mapping of design spaces from which molecular design rules and functional candidates can emerge. Current efforts are focused on elucidating the molecular origin of the concentration-dependent gelation behavior of **P11** in DME/DOL-based electrolytes and its influence on electrolyte organization. Beyond the present system, this modular molecular editing strategy offers opportunities for extension to commodity and waste-derived polymers (such as vulcanized rubber), enabling the generation of functional polymer libraries and potentially accelerating the discovery of advanced materials for energy storage, separations, and circular polymer technologies.

Experimental procedures

Details regarding the experimental procedures can be found in the supplemental experimental procedures.

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Hatice Mutlu (hatice.mutlu@uni-jena.de).

Materials availability

All reagents in this study are commercially available or can easily be synthesized. Full experimental procedures as well as spectra are provided in the supplemental information.

Data and code availability

Additional data and code for the TEM image analysis are available on request.

Acknowledgements

This work was conducted as a joint collaborative project, and the authors gratefully acknowledge this joint effort. H. M. acknowledges the University of Haute-Alsace for the financial support from the French National Research Agency with the reference “ANR-22-CPJ1-0077-01” for covering the PhD studies of J. J., the Department of Chemistry, RPTU University Kaiserslautern-Landau, and the support from the Ministry of Science of Rhineland-Palatinate. H. M. and S. D. acknowledge the Eucor Seed Money project “FuseTheWaste,” which supported the collaboration underlying this work. O. E. E. acknowledges the Petroleum Technology Development Fund (PTDF) and the German Academic Exchange Service (DAAD) under Nigerian-German Postgraduate Training Programme PhD, 2024 (57702733) for financial support covering his doctoral studies at University of Freiburg, Germany, Department of Sustainable Systems Engineering. I.M acknowledges the Ceramics & Composites Laboratory of the University of Ioannina and A.

Kaloudi for the excellent support regarding the AFM measurements. A. K. gratefully acknowledges the Verband der Chemischen Industrie (VCI) for financial support through a Liebig Fellowship. The authors thank the Karlsruhe Institute of Technology (KIT) for providing access to the analytical facilities used for SEC measurements in DMAc; and Kuraray Europe GmbH for the kind donation of PBD.

Author contributions

J.J.: conceptualization, methodology, investigation, formal analysis, data curation, visualization, writing - original draft of the whole text, and writing - review and editing; polymer synthesis, purification, and isolation of **P1-P18**; NMR spectroscopy; SEC (THF); FTIR spectroscopy; TGA; DSC; TEM sample preparation, qNMR analysis for functionalization degree, low-field ^{1}H NMR data treatment, quantitative sensitivity analysis. O.E.E.: investigation, formal analysis, visualization, and writing – original draft (Li–S electrochemistry section); writing – review and editing; design and execution of the electrochemical evaluation of **P11** as an electrolyte additive for Li–S batteries, including cyclic voltammetry, electrochemical impedance spectroscopy, galvanostatic cycling, symmetric-cell (S‖S, Li‖Li) measurements, and electrolyte-volume/concentration screening. C.V.: low-field ^{1}H NMR T1 measurements. A.K.: SEC measurements in DMAc. A.S.: supervision (Li–S electrochemistry), formal analysis, and writing – review and editing; contribution to data interpretation and discussion of trends in the Li–S battery study. S.D.: supervision of O.E.E., resources, and writing - review and editing; contribution to the interpretation and discussion of the Li-S electrochemical data. H.M.: conceptualization, methodology, supervision, project administration, funding acquisition, and writing - review and editing.

Declaration of interests

The authors declare no competing interests.

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